

MP Conversation Framework

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Intro

This document describes a system for classifying conversations so that Alexa and other multimodal experiences can determine *when* and *what* to show (primary and secondary fragments) and *how* to guide conversations across voice and visual interfaces. The goal is to create a shared language for everyone that aligns conversation structure, design intent, and model behavior.

Having a consistent classification framework allows us to establish CX principles and design guidelines upfront to help shape the LLM and not solely rely on parametric constraint solvers/reinforcement learning alone to refine the experience. By structuring conversations into identifiable modes and steps, we can better coordinate between spoken and visual responses, improve model predictability, and deliver experiences that feel coherent, adaptive, and purpose-driven across contexts.

A catalog of ten Conversation Types is included for scenarios and pattern reviews; these are exemplars used to apply the framework, not a taxonomy.

Conversational Step Taxonomy

This taxonomy outlines the core moments that make up nearly every human conversation. It draws from research in conversation analysis and dialogue theory (Sacks, Schegloff, & Jefferson, 1974; Clark & Brennan, 1991; Stolcke et al., 2000) and provides a shared vocabulary for understanding the rhythm of interaction. Conversational steps are the operative states used to design, tag, and evaluate turns: each step captures the user's immediate cognitive goal, such as clarifying, learning, comparing, or deciding, and the agent's corresponding role at that moment. Applied per turn, steps coordinate voice and visual behavior, shape prompt logic, and provide a consistent labeling scheme for analytics and quality review. They are modular and recurrent rather than a fixed script, and they can repeat, loop, overlap, or be skipped as context shifts, with a single prevailing step typically associated with each turn for clarity. In tandem, behavior patterns describe how information is engaged, and fragments express the state visually, keeping the interaction coherent.

	#	Conversational Step	User's Cognitive Goal	Agent Strategy
1	1	Opening / Initiation	Begin or re-engage in dialogue. Establish readiness or intent to interact.	Establish presence, acknowledge the user, and set context.
2	2	Clarify / Elicit	Reduce ambiguity or fill in missing details.	Ask targeted questions, rephrase, or confirm understanding
3	3	Inform / Provide	Obtain or share information relevant to the goal.	Deliver concise, relevant content and confirm receipt.
4	4	Explore / Elaborate	Examine related or expanded information.	Offer options, related ideas, or deeper details based on user curiosity.
5	5	Evaluate / Compare	Weigh trade-offs or compare possibilities.	Summarize differences, show pros and cons, and help the user make sense of alternatives.
6	6	Decide / Commit	Make a choice or confirm a direction.	Validate decision, confirm intent, and prepare to act.
7	7	Execute / Outcome	See a result or complete an action.	Perform the task, show status, and report success or issue.
8	8	Close / Reflect	End the exchange or transition gracefully.	Acknowledge completion, express thanks, or prompt next need.

These steps are not rigid stages but flexible conversational states that can appear, loop, or overlap.

Fragment Typology

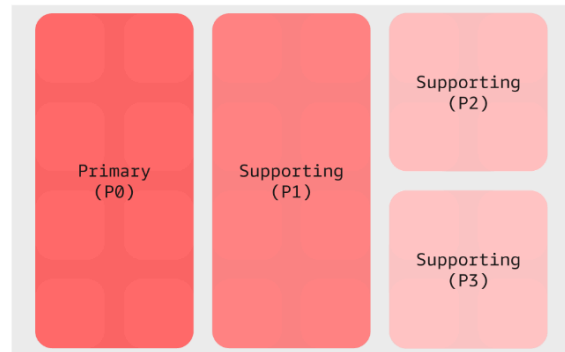
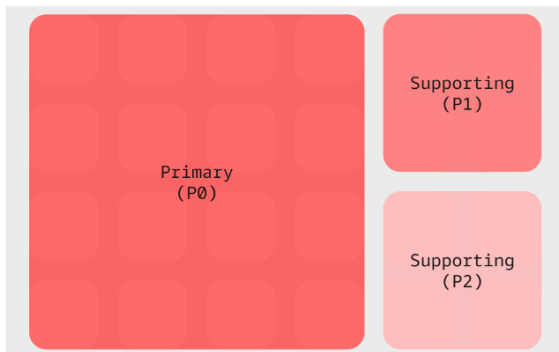
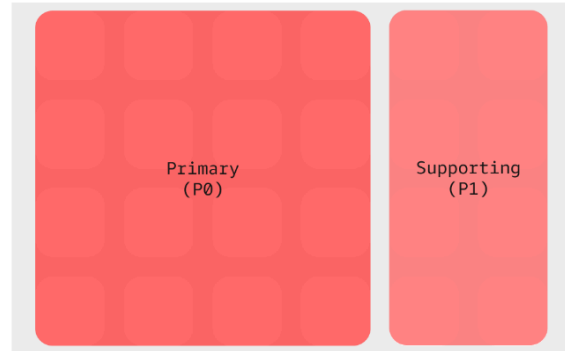
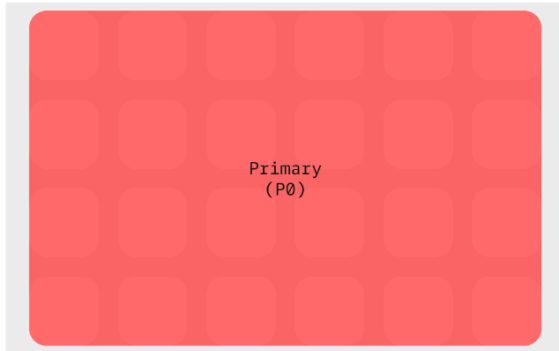
To support visual and verbal experiences that remain clear and contextually relevant, this model defines two complementary layers: primary fragments and supporting fragments.

Primary fragments carry the core content or interaction for the current conversational turn.

They represent the main focus of user intent such as viewing a list of results, completing a form, or confirming an action. Primary fragments establish the central task and remain the user's anchor throughout that exchange.

Supporting fragments are secondary elements that appear alongside the primary content when additional context or guidance improves the experience. Their role is to clarify, assist, or provide situational awareness without introducing distraction or redundancy. Supporting fragments adapt to the user's goal at that moment and can be classified by the outcome they enable.

In most scenarios, the majority of the TTS payload will support the primary fragment. The TTS can provide the key data points or essence of the supporting fragments if it meets a helpfulness threshold. This will also impact headless devices.



Example Scenario: Bike Trail

User Goal: Go trail riding this weekend.

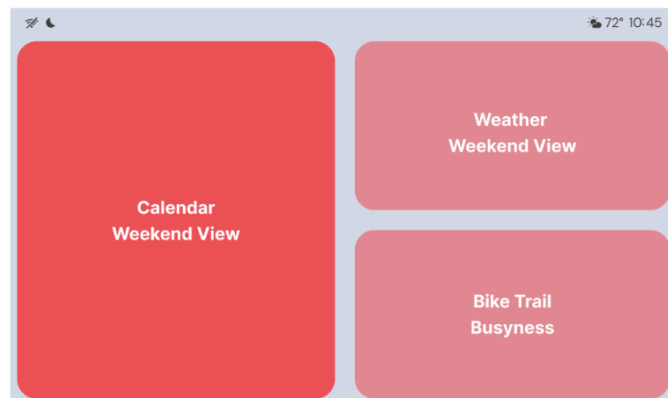
Current User Task: Choosing what day to go biking.

TTS: "Saturday is wide open, you have a birthday party on Sunday. Which day were you thinking?"

Primary: See agenda all weekend to help choose day.

Support 1: Weekend weather to help choose day.

Support 2: Bike Trail popularity to help choose day.



Outcome-Driven Supporting Fragments

Supporting fragments can be grouped into four outcome categories that describe their purpose within the conversation.

1. **Learn** adds context or explanation to improve understanding. Examples include definitions, diagrams, or reference visuals.
2. **Decide** aids evaluation or comparison through filters, key metrics, or summaries that help narrow choices.

3. **Execute** supports task completion through function-focused visuals such as controls, progress, or confirmation indicators.
4. **Engage** maintains continuity or rapport through expressive or ambient visuals that reinforce tone and personality.

This outcome-driven approach focuses on *why* supporting content appears rather than *what* specific form it takes. Aligning visuals to user intent ensures that additional information increases clarity rather than cognitive load.

Behavior Patterns

Behavior Patterns describe how users interact with information. They define the concrete ways users consume, navigate, and manipulate content regardless of the broader conversation goal. These patterns are domain-agnostic, which means the same pattern applies whether someone is shopping for products, choosing restaurants, or browsing entertainment. They form the bridge between conversational purpose and presentation, allowing the design system to scale through reusable layouts and logic.

	#	Behavior Pattern	Description	Typical UI Pattern	Example Interaction
1	1	Glancing	Viewing a single piece of ephemeral information.	Minimal card, auto-dismiss	"What time is it?"
2	2	Reviewing List	Scanning multiple items at the same level.	List or grid layout	"Show nearby coffee shops."
3	3	Comparing Items	Evaluating two to four options side by side.	Comparison table or aligned cards	"Which laptop is better."
4	4	Inspecting Single Item	Exploring details of one item deeply.	Detail card with rich content	"Tell me more about the second restaurant."
5	5	Filtering / Narrowing	Refining results through constraints.	Filters, sliders, active chips	"Show only Italian restaurants under \$30."
6	6	Confirming	Approving or validating a single action.	Confirm or cancel dialog	"Place my order."
7	7	Composing	Creating or editing something incrementally.	Step-by-step form or builder	"Add eggs to my shopping list."
8	8	Monitoring Active State	Observing live or changing information.	Persistent card or progress bar	"Play the new Ghostbusters movie"
9	9	Following Sequence	Progressing through ordered steps.	Stepper or progress tracker	"Walk me through setting up my router."
10	10	Exploring Tangent	Following a related but new topic or path.	Branching card or breadcrumb	"Show me other recipes like this one."

Integration: How the Layers Work Together

This framework works by aligning three layers that operate in parallel: Steps, Behavior Patterns, and Fragments. Each plays a distinct role, but together they create a unified logic for how conversations unfold across voice and visual experiences.

The Conversation Step layer defines the conversational moment—what’s happening cognitively and functionally. It describes the user’s goal at that point (clarifying, deciding, completing) and the agent’s corresponding role. This layer provides structure to the flow of interaction and is what allows both designers and the LLM to recognize “where” they are in the conversation.

The Behavior Pattern layer defines how the user is engaging with information at that moment. A user might be scanning a list, inspecting an item, or confirming a choice. These patterns inform how the interface should behave—how dense information should be, how interaction elements are arranged, and how the agent's voice pacing should adjust to support that behavior.

The Fragment layer defines what is shown or expressed in that moment. Primary fragments anchor the user's attention to the core task or message. Supporting fragments add context or help accelerate decision-making when useful. The fragment system ensures that the screen and voice reinforce one another, instead of competing for focus.

When these layers work together, they form a single synchronized system. The Step defines intent, the Behavior Pattern defines interaction, and the Fragment defines presentation. This alignment keeps the voice and visuals in lockstep, letting each modality express the same state through different channels.

By treating these layers as interconnected rather than sequential, the framework enables consistent, adaptive experiences that scale across domains and devices. The LLM can identify conversational state and respond accordingly, while designers can ensure every moment feels coherent and contextually aware.

Conversation Types (for Examples)

The list below defines ten Conversation Types we will use to author examples and review patterns. These are exemplars, not a taxonomy. They sit on top of the Conversation Framework above and help us select realistic scenarios that exercise the Steps, Behavior Patterns, and Fragments. Each type tends to emphasize a characteristic step path.

1. Closed-Ended Q&A

A short factual exchange aimed at a single, unambiguous answer with minimal follow-up.

Example: "What is $10 + 10$?"

Typical step path: Opening → Clarify (if needed) → Inform → Close

2. Open-Ended Information

Exploratory information seeking where scope evolves as the user learns and narrows what matters.

Example: "What should I know before visiting Japan for the first time?"

Typical step path: Clarify → Inform → Explore → Inform → Close

3. Proactive Notification & Follow-Up

An agent-initiated alert that communicates status and offers an optional next step when relevant.

Example: "Your package is delayed until tomorrow—do you want delivery alerts?"

Typical step path: Opening (system) → Inform → Decide → Execute → Close

4. Quick Action

A one-shot command where immediate execution and confirmation matter more than exploration.

Example: "Turn on the porch light."

Typical step path: Opening → Execute → Close

5. Reasoning & Comparison

Choosing among known options using explicit criteria supplied or confirmed by the user.

Example: "Compare these two 13-inch laptops for school"

Typical step path: Clarify → Inform (results) → Evaluate (compare) → Decide → Execute → Close

6. Complex / Chain-of-Thought

Deriving an answer through multi-step reasoning, assumption checks, or decomposition when the result isn't directly available.

Example: "Which laptop under \$1,200 gives me the best color accuracy for photo work over the next four years?"

Typical step path: Clarify → Inform/Evaluate (micro-loops) → Decide → Execute → Close

7. Troubleshooting & Resolution

Diagnosing a problem, proposing remedies, and verifying the fix through targeted checks.

Example: "My Wi-Fi keeps dropping."

Typical step path: Opening → Clarify (scope/symptoms) → Inform (tests) → Evaluate (likely cause) → Decide (remedy) → Execute → Close (verify)

8. Planning & Advisory

Co-creating a concrete plan with goals, constraints, options, and accountability over time.

Example: "Help me plan my wedding."

Typical step path: Clarify (goals/constraints) → Explore (options) → Evaluate (fit) → Decide (plan) → Execute (schedule/reminders) → Close (next check-in)

9. Motivational & Progress

Encouraging behavior change and sustaining momentum with light commitments and periodic check-ins.

Example: "Keep me on track to drink more water every day."

Typical step path: Opening → Clarify (goal/barriers) → Inform (tips) → Decide (commitment) → Execute (reminders/progress) → Close

10. Chit Chat / Social

Rapport building for presence and delight without a task-heavy outcome.

Example: "Tell me a joke."

Typical step path: Opening → Inform/Explore (light content) → Close

References

Clark, H. H., & Brennan, S. E. (1991). *Grounding in communication*. In Resnick et al. (Eds.), *Perspectives on Socially Shared Cognition*.

Sacks, H., Schegloff, E. A., & Jefferson, G. (1974). *A simplest systematics for the organization of turn-taking for conversation*. *Language*.

Stolcke, A., et al. (2000). *Dialogue Act Modeling for Automatic Tagging and Recognition of Conversational Speech*. *Computational Linguistics*.

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Hassenzahl, M., & Tractinsky, N. (2006). *User Experience – a research agenda*. *Behaviour & Information Technology*.

Miller, G. A. (1956). *The Magical Number Seven, Plus or Minus Two*. *Psychological Review*.

Related Docs

- [12 Conversational Archetypes for Human-AI Interaction](#) (full research report)
- [Moneypenny BRD](#)
- [Alexa Conversation Types](#)
- [D&S Customer Needs Framework](#)
- 12 Archetypes of Conversation: Design Reference
- [Sample Conversations for Moneypenny with Visuals](#)

Change Log

- 10/13/2025 - Added Conversation types as a hybrid of the 12 Conversation Archetypes and the existing canonical types used internally
- 10/13/2025 - Removed Conversation Modes to reduce classification systems.
- 10/13/2025 - Moved from 12 Conversational Archetypes to an aggregation of the steps within all conversation types (Conversation Steps).

DO NOT READ PAST HERE

Archive

Conversation Modes

Conversation mode describes the overall tone, pacing, and purpose of an interaction. The difference between a social chat and a task-driven exchange. Most human conversations can be viewed through two primary modes, and people fluidly move between them.

Casual Mode: Open and social. The goal is rapport or shared experience, not efficiency. The tone is light, expressive, and sometimes playful.

Transactional Mode: Focused and purposeful. The goal is progress—completing a task, confirming a fact, or making a decision. The tone is concise and confident. Visuals lean utilitarian, favoring clarity and speed.

A single interaction can shift between these modes. The framework is designed to accommodate that fluidity: a conversation can start casually (“Alexa, tell me a joke”) and naturally move into a transaction (“Actually, can you add coffee to my order?”) without breaking continuity.

Conversation Stages

Each conversational archetype follows a recognizable sequence of stages that describe how the exchange naturally unfolds.

Stages represent the rhythm or flow of the conversation, from how it begins, through how information or action is exchanged, to how it closes. Most archetypes share three broad phases: **Opening**, **Business**, and **Closing**.

Within these phases, each archetype expresses its own specific sub-stages such as defining goals, exploring options, or confirming choices. Stages are not rigid steps but recurring patterns that help identify the user’s focus and determine which visual fragments are most appropriate at each point in the interaction.